

Review for

“Minomaly: A Subgraph Mining Framework for Unsupervised and Interpretable Anomaly”

Summary of the paper

This paper proposes Minomaly, an unsupervised and interpretable framework for structural graph anomaly detection. The method consists of two main stages. First, it constructs an Order Embedding Space (OES) using a GraphSAGE-based GNN to encode subgraph containment relations and approximate the frequency of local graph structures. Second, it performs an anomalous-structure search by greedily expanding low-frequency node neighborhoods based on OES geometry, and identifies anomalous nodes through the detection of rare structural patterns. The paper also presents a theoretical argument based on the anti-monotonicity of the proposed frequency measure and reports experiments on Cora, Amazon, and Flickr.

Main strengths

1. The overall pipeline is technically coherent. The paper presents a complete framework, from OES construction and frequency estimation to greedy anomalous-structure search.
2. The interpretability goal is meaningful. Unlike many existing GAD methods that only return anomaly scores or classification results, Minomaly tries to identify the anomalous local structures behind the node-level decisions. This is practically valuable.
3. The empirical results are promising on the selected baselines. On the chosen comparison set, the method performs well across datasets of different sizes. The Flickr results also suggest good scalability compared with the selected baselines.

Main weaknesses

1. The novelty positioning is the main issue. The manuscript states that SPMiner “relies on order embedding to count frequent graph patterns,” and later says that the implementation “relies on SPMiner [28] for subgraph mining.” However, the paper does not explain clearly enough what is inherited from earlier order-embedding/subgraph-mining work, what is adapted for anomaly detection, and what is new in Minomaly.
2. The related work section is not well organized and does not clearly show the gap addressed by the paper. It mixes several topics, including general anomaly detection, graph anomaly detection, explainability, pattern mining, and order embedding. However, it does not build a clear path from

prior work to the specific problem studied here. As a result, the section does not give a clear research context.

3. The literature framing should be more careful. The repeated use of “state-of-the-art” is not well supported in the current version. The main baselines are GCNAE (2016), DOMINANT (2019), AnomalyDAE (2020), DONE (2020), AdONE (2020), and CONAD (2022). Since most of them are from the 2019–2022 period, the repeated use of “state-of-the-art” seems too strong for the current literature coverage.
4. The academic presentation should be improved, especially in the related work and novelty discussion. For example, the discussion of SPMiner includes a sentence that is too long and hard to parse: “Among these, SPMiner relies on order embedding [30, 31] to count frequent graph patterns, yet it lacks accuracy in predicting graph-level frequency as this was its primary focus to estimate it, and the correctness of frequency estimates based on exact motif matching was not proven.” This kind of writing makes the comparison harder to follow and weakens the paper’s novelty positioning.

Minor issues

I also noticed several editorial issues:

- Page 2: “Graph a Anomaly Detection (GAD) plays a crucial role in uncovering hidden insights and mitigating risks in various domains.”
- Reference [48]: “Erd6s” should be corrected to “Erdős”.

Suggestions for revision

A future revision would benefit from the following changes:

1. Add a short discussion that explains more clearly how Minomaly differs from earlier order-embedding/subgraph-mining work. This discussion should state which parts are reused and which parts are specific to anomaly detection.
2. Add more recent work to the related work section. Also reorganize this section into clearer subsections and end it with a short gap statement. For example, the section could first discuss graph anomaly detection, then interpretable anomaly detection, and then the order-embedding/subgraph-mining line most related to this paper.
3. Reconsider the repeated use of “state-of-the-art” in the abstract and experimental discussion. If the authors want to keep this wording, they should provide stronger support, for example by adding more recent baselines or by explaining more clearly why the current baselines are enough to support such a claim.

4. Revise the manuscript for clarity. In several places, especially in the related work and technical comparison sections, some sentences are too long and too complex. Shorter and simpler sentences would improve readability.

Overall recommendation

While I see publication potential in the core idea, the problems in the current manuscript are too substantial to be fully addressed within a standard major-revision cycle. For this reason, I recommend rejection in the current form.